Experiment-1

Basic networking command in Linux and windows

Aim:

To study and execute basic networking commands available in Linux and windows os for diagnosing and troubleshooting network issues.

Requirements:

PC with window / Linux, terminal or command Prompt Active network connection.

Procedure:

1) Open command Prompt or terminal

2) Execute 'ipconfig' or 'ifconfig' to view IP configuration.

3) use Ping 8.8.8.8 to test connectivity to google DNS server.

4) execute 'tracert google.com' or 'traceroute google.com' to view Path Packet.

5) Run netstat -an to list active network connections and listening Ports.

Commands

1) ipconfig/all # Windows full ip configuration.

2) ip a # Linux ip address info

3) Ping -c4 8.8.8.8 # Linux ping 4 packets.

4) Ping 8.8.8.8 # Windows Ping

6) netstat -tuln # Linux Listen Port.

7) nslookup google.com # DNS resolution

8) arp -a # ARP table

9) route -n # Routing table (Linux).

6) use 'nslookup google.com' to perform DNS lookup

7) Record all output and analyse the results.

Result: 8/10

24/9/26. Basic networking commands were successfully executed on both Linux and windows. IP address, routing tables, ARP caches and DNS resolution were observed and recorded.